

SharePoint Integration API (.NET 10)

This API integrates with Microsoft Graph using **Microsoft.Graph SDK v4.54** and client-credential authentication for SharePoint/OneDrive operations.

Technologies

- .NET 10.0
- Microsoft.Graph 4.54.0
- Microsoft.Identity.Client 4.66.0
- ASP.NET Core Web API
- Swagger/OpenAPI

Configure

Update the `Graph` section in `appsettings.json` or `appsettings.Development.json`:

```
{
  "Graph": {
    "TenantId": "your-tenant-id",
    "ClientId": "your-client-id",
    "ClientSecret": "your-client-secret",
    "Scopes": ["https://graph.microsoft.com/.default"]
  }
}
```

Azure App Registration Steps

1. Go to **Azure Portal** → **Microsoft Entra ID** → **App registrations** → **New registration**
2. Name the app (e.g., `SharePointIntegrationApi`) and select **Single tenant**
3. Click **Register**
4. Copy **Application (client) ID** and **Directory (tenant) ID** from the overview page
5. Create a secret: **Certificates & secrets** → **New client secret** → copy the secret value
6. Add permissions: **API permissions** → **Add a permission** → **Microsoft Graph** → **Application permissions**
7. Add required permissions:
 - `User.Read.All` - Read all users' profiles
 - `Sites.ReadWrite.All` - Read and write to all SharePoint sites
 - `Files.ReadWrite.All` - Read and write files in all site collections
8. Click **Grant admin consent for [your tenant]**

SharePoint Online License Requirements

Drive endpoints require SharePoint Online licensing:

1. In the **Microsoft 365 admin center**, go to **Billing** → **Licenses** and verify a plan that includes **SharePoint Online**
2. Assign the license to at least one user (the user configured in `userId`)
3. Ensure the user's **OneDrive** is provisioned (first sign-in or provision via admin center)
4. Confirm app permissions include `Sites.ReadWrite.All` and `Files.ReadWrite.All` with admin consent

API Endpoints

User Information

- **GET** `/graph/users` - Get the configured user's profile (ID, display name, UPN)

Drive Root Operations

- **GET** `/graph/users/drive/root` - Get the user's OneDrive root folder information
- **GET** `/graph/users/drive/root/folders` - List all folders in the user's OneDrive root

Folder Management

- **POST** `/graph/users/drive/root/folders?folderName={name}` - Create a new folder in the user's OneDrive root

- Automatically renames if folder exists
- Returns folder ID, name, and web URL

File Upload

- **POST** /graph/users/drive/root/folders/upload?folderPath={path} - Upload a file to a specific folder
 - **Query Parameters:**
 - folderPath: Target folder path (e.g., "Reports/2024")
 - **Body:** Form-data with file
 - **Constraints:**
 - Max file size: 50 MB
 - Allowed extensions: .doc, .docx, .xls, .xlsx, .pdf, .eml, .msg
 - Returns uploaded file ID, name, web URL, and size

Secure Folder (Advanced Permission Management)

- **POST** /graph/users/drive/root/folders/secure - Create a folder with restricted access
 - **Request Body:**

```
{
  "folderName": "Confidential Reports",
  "folderPath": "Projects/2024",
  "recipients": ["user1@domain.com", "user2@domain.com"],
  "role": "read",
  "removeExistingPermissions": true,
  "message": "You have been granted access to this folder"
}
```
 - **Parameters:**
 - folderName (required): Name of the folder to create
 - folderPath (optional): Parent path where folder will be created
 - recipients (required): Array of email addresses to grant access to
 - role (optional): Either "read" or "write" (defaults to "read")
 - removeExistingPermissions (optional): Remove all non-owner permissions before adding new ones (defaults to true)
 - message (optional): Custom message to include with the permission grant

Permission Management

- **POST** /graph/users/drive/items/{itemId}/permissions - Add permissions to an existing folder/file
 - **Path Parameters:**
 - itemId: The ID of the folder or file
 - **Request Body:**

```
{
  "recipients": ["user@domain.com"],
  "role": "read",
  "message": "Optional message"
}
```
 - **Parameters:**
 - recipients (required): Array of email addresses
 - role (optional): "read" or "write" (defaults to "read")
 - message (optional): Custom message

How Secure Folder Works

The secure folder endpoint (POST /graph/users/drive/root/folders/secure) provides advanced permission management:

Process Flow

1. Check for Existing Folder

- First checks if the folder already exists at the specified path
- If it exists and is a folder, uses the existing folder
- If it doesn't exist, creates a new folder with conflict behavior set to "fail" to prevent duplicates

2. Remove Existing Permissions (if `removeExistingPermissions` is true)

- Lists all current permissions on the folder
- Identifies which users are in the allowed recipients list
- Adds the folder owner to the allowed list to prevent removing their access
- Removes all permissions that don't belong to the owner or specified recipients
- This ensures ONLY the specified users have access

3. Grant New Permissions

- Sends invitations to all specified recipients
- Grants either "read" or "write" access based on the role parameter
- Uses Microsoft Graph's invite API to properly set permissions
- Does not send email notifications (`sendInvitation: false`)

Permission Levels

- **read:** Recipients can view and download files
- **write:** Recipients can view, download, upload, and modify files

Security Features

- **Explicit Permission Control:** When `removeExistingPermissions` is true, only specified recipients and the owner retain access
- **No Duplicate Folders:** Uses conflict behavior "fail" to prevent creating duplicate folders
- **Owner Protection:** Automatically preserves the folder owner's permissions
- **Input Validation:** Validates email addresses and roles before processing
- **Error Handling:** Provides detailed error messages for troubleshooting

Example Usage

Create a folder only accessible by specific users:

```
POST /graph/users/drive/root/folders/secure
Content-Type: application/json

{
  "folderName": "Q4 Financial Reports",
  "recipients": ["cfo@company.com", "accountant@company.com"],
  "role": "read",
  "removeExistingPermissions": true
}
```

Manual Alternative

If you prefer to manage permissions manually using Graph API:

1. **Create the folder:** POST `/users/{userId}/drive/root/children`
2. **List current permissions:** GET `/users/{userId}/drive/items/{itemId}/permissions`
3. **Remove unwanted permissions:** DELETE `/users/{userId}/drive/items/{itemId}/permissions/{permissionId}`
4. **Invite specific users:** POST `/users/{userId}/drive/items/{itemId}/invite`

Required Microsoft Graph Permissions

Application Permissions (requires admin consent):

- `User.Read.All` - Read user profiles to resolve email addresses
- `Files.ReadWrite.All` - Full read/write access to files
- `Sites.ReadWrite.All` - Full read/write access to SharePoint sites

Swagger Documentation

Swagger UI is available at `/swagger` when the application is running. Provides interactive API documentation and testing interface.

Running the Application

```
# Restore packages
dotnet restore

# Build the project
dotnet build

# Run the application
dotnet run
```

The API will be available at <https://localhost:5001> or <http://localhost:5000>.

Notes

- Uses confidential client with client credentials flow (app-only authentication)
- All operations are performed in the context of the configured user
- Requires proper Azure AD app permissions with admin consent
- OneDrive must be provisioned for the configured user
- File size limit for uploads is 50 MB (configurable in code)